Start with a working bench

A practical wiring and power-up guide for your sensor-based plant-care robot. Start with USB and individual sensors. Add actuators only after their ratings and drivers are identified. Battery wiring comes last.

Known from your photos: Uno R3, DFRobot FireBeetle ESP32 V4.0, Raspberry Pi Pico W, webcam, two IR modules, ultrasonic module, soil probe and interface, pump, tubing and chassis. **Unconfirmed:** exact sensor variants/pin order, MSP430 model, motor driver, pump-switch model, motor/pump ratings and battery lead functions.

Where	Planned role
Fixed station	Laptop + webcam; MSP430 LaunchPad + course Booster Board; FireBeetle ESP32 if needed as the USB/UART/Wi-Fi gateway.
Moving robot	Pico W; two IR sensors; ultrasonic sensor; dual wheel driver; pump switch; tank; regulated battery supply.
Uno R3	Temporary sensor tester. It does not have to remain in the finished design.
One test pot	Keep the moisture probe in this pot. Final analog input goes to the fixed MSP430 after its pinout is known.

1. Gather the small parts

Use a multimeter, breadboard, male/female jumper wires, USB-B data cable for the Uno, micro-USB data cables for the Pico/ESP32, black tape and a light floor sample. For the ultrasonic divider, get one 10 kOhm and one 15 kOhm resistor (1% preferred). A ruler and measuring cup are useful for calibration.

For later power wiring: correctly rated dual motor driver, pump driver, regulators, fuse/holder, switch, terminal connectors, insulated stranded wire and the battery manufacturer-matched charger. Their current ratings remain to be selected.

2. Make the work area readable

Label rails **GND**, **3V3** and **5V**. A breadboard colour stripe does not create a voltage; only the wire feeding it does. Rail segments can be split: check continuity with all power disconnected. Mount the Pico across the centre gap so opposite pins are not shorted together.

Power-off rule: unplug USB and disconnect battery/bench supplies before changing wires. Keep motor and pump current out of breadboard rails and thin signal jumpers. Use a dry tray for water tests and keep water away from the electronics.

Power up each board separately

Leave sensors, drivers and inter-board wires disconnected for these first checks. USB power tests do not require a robot battery.

Item	Action	Pass condition
Uno R3	Connect its USB-B socket to the laptop. In Arduino IDE select Arduino Uno and its port. Open the supplied uno_sensor_bench sketch with TEST_MODE = 0; upload. Open Serial Monitor at 115200 baud.	Repeated heartbeat=OK messages; built-in LED toggles.
Pico W	Hold BOOTSEL while connecting micro-USB, then release it. The bootloader should appear as RPI-RP2. This identifies the bootloader; it is not the final program.	RPI-RP2 drive appears. A dark LED alone does not indicate a fault.
FireBeetle ESP32 V4.0	Connect micro-USB. Use the DFRobot DFR0478 getting-started instructions for its board package and CH340 driver if needed. Upload a simple serial test and check its port.	The expected serial message appears; no resets when idle.
Webcam	Connect directly to laptop USB. Open Windows Camera, select the USB camera, and check the view of the pot. Close Camera before another app uses it.	Clear, stable image of the intended pot.
MSP430 + Booster	Identify the exact LaunchPad first. Follow your lab orientation/jumper instructions for fitting the Booster with power removed. Use the course IDE and its original lab blink/serial test.	The known lab example runs. No external sensor wiring yet.

3. Prepare Pico C/C++ development

Use the official Raspberry Pi Pico VS Code extension. Create a C/C++ project for **Pico W / RP2040**, enable USB stdio, build a basic serial hello-world, and copy its UF2 to RPI-RP2. Reopen the serial port after the board restarts. Selecting plain Pico or using GP25 as the Pico W LED can give misleading results. [R1, R2]

Voltage check: put the meter black lead in COM and red lead in V/Ohm; select DC volts. With USB connected, measure Uno 5V-to-GND and Pico 3V3(OUT)-to-GND. Expect about 5 V and 3.3 V. Never put an ammeter directly across a supply.

Keep the FireBeetle white battery socket empty in this shared-pack design. It is intended for a single-cell 3.7 V lithium battery, not a 7.4 V pack. [R3]

Read the sensors with the Uno

These are **USB-only bench tests**. Test one mode at a time. The exact pin order varies between modules: follow printed VCC/GND/OUT/AO labels, not the left-to-right order in a generic photo. If labels cannot be identified, leave that module disconnected.

Test mode	Module connection	Uno connection
1: soil moisture	VCC on a module confirmed to support 5 V	5V
	GND / AO (analog output)	GND / A0
	Two-prong probe	Its matching two-terminal interface connection; DO unused
2: IR pair	Each VCC / each GND, for 5 V-capable modules	5V / GND
	Left OUT / right OUT	D2 / D3
3: standard HC-SR04	VCC / GND	5V / GND
	TRIG / ECHO	D4 / D5, direct on the 5 V Uno only

4. Soil probe

With USB removed, wire the probe/interface. Set TEST_MODE = 1, reconnect and upload. Read soil_adc at 115200 baud. Record dry and damp samples at the same insertion depth; wait for each reading to settle. Keep the small interface board above the soil. Do not assume a higher number means wetter: verify the direction experimentally.

The Uno uses a nominal 5 V ADC reference by default; raw counts range from 0 to 1023. They are not moisture percent. Unplug after tests; continuous power can accelerate corrosion of exposed resistive probes.

5. IR pair

Set TEST_MODE = 2. Aim both modules downwards and move black tape and the light surface underneath. Adjust the trim pots and mounting height until both distinguish the surfaces repeatedly. Write down which output means black. These appear to be proximity-style modules: they must pass this test before being accepted as line sensors. [R5]

6. Ultrasonic sensor

Set TEST_MODE = 3. Place a flat target at measured distances such as 10, 20 and 40 cm; compare the reported values with a ruler. NO_ECHO means an invalid measurement, not a clear route. The sketch waits between pings and limits its echo wait. [R4]

Pass before moving on: clear dry/damp separation; reliable black/light readings on BOTH IR sensors; sensible distance readings. This sketch uses blocking delays for testing and does not fulfil the final timers/ISRs/control requirements.

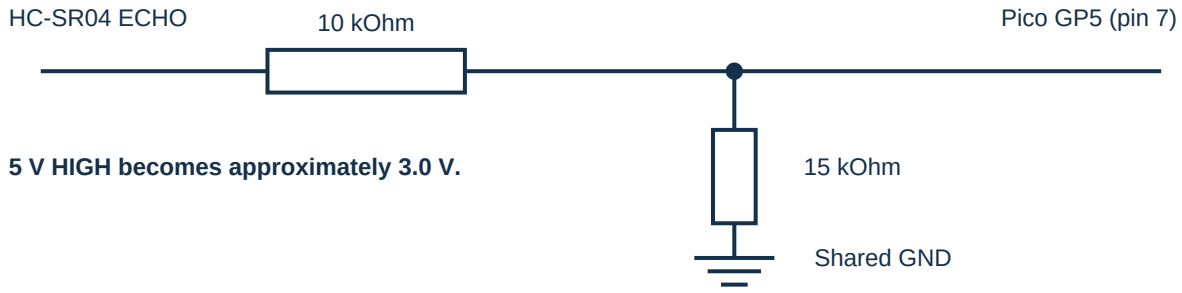
Move navigation sensors to Pico W

Disconnect their Uno wires completely before moving them. Below is a **proposed pin assignment**, not a claim that these wires are already fitted. The soil sensor stays at the fixed pot in the final system.

Connection	Pico signal	Physical pin
Left IR OUT (0-3.3 V only)	GP2	4
Right IR OUT (0-3.3 V only)	GP3	5
IR VCC, if rated for 3.3 V	3V3(OUT)	36
IR/ultrasonic ground	GND	38, or another GND
HC-SR04 trigger	GP4	6
HC-SR04 echo via divider below	GP5	7
HC-SR04 VCC during Pico USB-only bench test	VBUS (USB 5 V)	40

Count carefully: with components facing you and USB at the top, physical pin 1 is top left; pin 20 bottom left; pin 21 bottom right; pin 40 top right. GP numbers are not physical pin numbers. Check the official pinout before wiring. [R1]

7. Build the ECHO voltage divider



Connect ECHO through 10 kOhm to a junction. Connect that junction to GP5 and through 15 kOhm to GND. The calculation is 5 x 15/(10+15) = 3.0 V. Before attaching GP5, test the divider using a steady 5 V input and a voltmeter. Remove that test input, then reconnect the sensor ECHO.

Never put 5 V on a Pico or ESP32 GPIO. The standard HC-SR04 uses 5 V power; its ECHO needs level reduction. GP4 gives a 3.3 V trigger: verify that your module responds; if it requires a higher trigger voltage, add an appropriate buffer rather than feeding 5 V into the Pico. [R1, R4]

Power the Pico by laptop USB for this stage. Do not inject external power into VBUS while laptop USB is connected. IR outputs must be checked in both states before connecting; if the module needs 5 V power, provide suitable output level conversion as well.

Add wheels and the water pump

Identification checkpoint: you have not confirmed the wheel-driver model, pump-switch model or actuator ratings. The connections on this page are conditional. Do not energise a motor or pump by guessing its voltage or the meaning of an unlabelled terminal.

8. Wheel driver: two independent motor channels

A pump switch is not a dual motor driver. Get a 3.3 V-compatible H-bridge whose supply range and continuous/starting-current ratings suit your two motors. A TB6612FNG carrier is one possible type only if its ratings are sufficient; do not buy it solely from the example below. [R7]

If using a verified TB6612FNG carrier	Proposed Pico connection
VCC / GND	3V3(OUT), pin 36 / shared GND
AIN1 / AIN2 / PWMA	GP6 pin 9 / GP7 pin 10 / GP8 pin 11
BIN1 / BIN2 / PWMB	GP9 pin 12 / GP10 pin 14 / GP11 pin 15
STBY	GP12 pin 16; hold LOW during startup; use a 10 kOhm pulldown if not provided
VM or VMOT	Motor-rated supply from its regulator, not Pico 3V3/5V
AO1 + AO2 / BO1 + BO2	Left motor pair / right motor pair

Use the carrier manufacturer's decoupling and wiring guidance. With wheels lifted clear of the table, test one wheel at low duty cycle, then the other, then reverse. If direction is wrong, remove power before swapping that motor pair. Keep STBY LOW until valid firmware is running. Do not add a single flyback diode across a bidirectional wheel motor.

9. Pump driver and tubing

If the black board is confirmed as DFRobot DFR0457, use its labelled VIN/GND input and VOUT/GND load terminals, and follow its S/+/- control labels. Its VIN minimum is 5 V; this must also suit the pump rating. It is not suitable simply because it resembles the photo. [R6]

For that confirmed module: GP13 (physical pin 17) is the proposed control signal, with shared signal GND and 3.3 V control supply according to the module pinout. Default OFF and a suitable pulldown are required. A flyback diode across the DC pump has its banded cathode at pump positive and anode at pump negative; choose ratings for the pump. Confirm whether the module already provides protection.

Secure the tank and tubing, identify inlet/outlet and follow the pump's priming/submersion instructions. Test short, supervised pulses into a measuring cup using a rated, current-limited supply. Record mL per pulse. Do not run an unknown pump dry or assume it is submersible.

Wire the fixed station

Keep the stationary boards on laptop USB or their documented regulated supplies. The robot battery only needs to ride on the robot. Wireless data does not transfer electrical power.

10. Fit the MSP430 and course Booster

Your manual requires an MSP430 LaunchPad and use of the CEG2002 Booster Board, plus at least one other microcontroller. The exact LaunchPad model and Booster schematic are needed to choose free ADC, UART and timer pins. Do not copy a pin number from a different LaunchPad. [R8]

Function	Connection once the board pinout is confirmed
Soil measurement	Sensor AO -> a free MSP430 ADC input. GND -> MSP430 GND. Use a sensor supply/output range compatible with that specific ADC and its reference; never assume 5 V is acceptable.
Approval / pause	Use an appropriate Booster button or interface already available in your lab setup. Confirm any shared pins before assigning an interrupt.
Status	Use an available Booster LED/display/buzzer as appropriate. This makes the Booster contribute to the application.
ESP32 gateway	MSP430 UART TX -> FireBeetle IO16 (configured RX); MSP430 UART RX <- FireBeetle IO17 (configured TX); GND <-> GND. Only after verifying compatible logic levels and free MSP430 pins.

11. Connect communications in small steps

First keep both fixed boards on their own USB supply. Join only UART TX/RX/GND between them; do not join their positive power rails. On the FireBeetle, IO16 and IO17 refer to the printed GPIO labels, not Arduino D16/D17. Configure the serial peripheral explicitly (for example, RX=16 and TX=17).

Prove UART using a counter and acknowledgment. Then prove ESP32-to-Pico W communication over ordinary 2.4 GHz Wi-Fi; do not assume ESP-NOW compatibility with the Pico W. A shared local access point or ESP32 access-point arrangement can be used after individual links work.

Keep motor and pump outputs inhibited while testing communication. Include a job identifier, fresh sensor status and acknowledgments. Repeated receipt of the same watering job must not repeat a dose. Loss of commands, invalid sensor data or a pause request must stop the affected outputs.

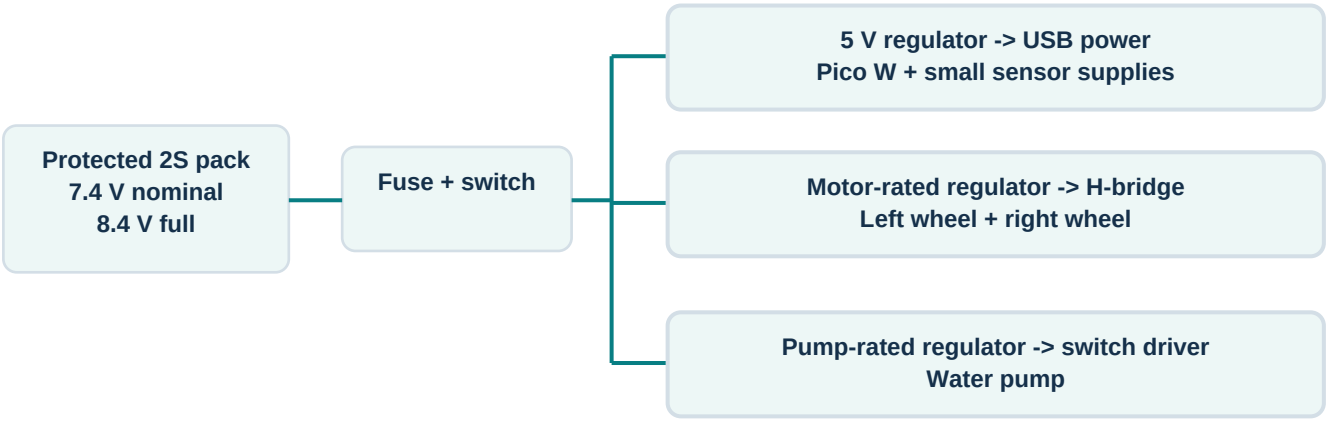
12. Camera and soil feedback

The webcam stays plugged into the laptop. Match each image and moisture reading to the test-pot ID. Begin with user approval of watering; a webcam alone does not diagnose plant disease. Use only one instrumented pot initially; additional independently controlled pots need their own valid moisture information.

The Uno test wiring is temporary. Do not leave the same analog output attached to differently powered boards while one is off. Move and recalibrate the soil sensor when its supply voltage or ADC reference changes.

Commission the battery supply last

The linked Unicell pack remains a shortlist. Its page lists 7.4 V, 2600 mAh, protection PCB and three wires, but not its allowable discharge current or the three-wire functions. Get those details and a compatible charger/connector from the supplier before connecting it. Do not assume the third wire is ground, a balance lead or a sensor lead. [R9]



13. Select the power components

Use an assembled protected pack and manufacturer-matched 2S lithium charger (typically 8.4 V CC/CV for this chemistry). The pack, protection, fuse, switch, connectors and regulators must support measured running and startup loads. A protection PCB is not a charger. The pack's mAh value does not establish its allowable current.

Choose the electronics regulator to maintain regulated 5 V across the chosen pack operating range. Choose wheel and pump supply voltages from their actual ratings. One actuator regulator can be shared only if both voltage requirements match and its current/transient capacity is sufficient. Never assume a fully charged 8.4 V pack can connect directly to 5 V or 6 V actuators.

14. Set voltages before connecting any board

With all loads disconnected, have the lab technician verify the pack's protected output pair and polarity. Connect pack positive through an appropriately selected fuse near the pack, then the main switch, to regulator IN+. Connect protected negative to IN-. If adjustable, set each output using a DC voltmeter, then disconnect power and secure the adjustment.

Confirm about 5.0 V at the electronics output. Have the supply checked under a representative test load. Label every output. Wire non-isolated regulator negatives, Pico GND and driver signal grounds to a common distribution return; route motor/pump return current directly there rather than through sensor ground wires.

15. Bring up the robot on battery

Disconnect laptop USB from the Pico. Feed the verified regulated 5 V into a correctly wired USB output socket and use a normal micro-USB cable to the Pico. Connect one load branch at a time: controller, sensors, wheels, then pump. Check for resets and voltage drops at each step. Keep the main switch reachable.

To program again, switch OFF and disconnect battery power before attaching laptop USB. Do not use a USB Y-cable to combine supplies. Charge with the pack disconnected from the robot, while attended, following the pack/charger instructions. Manual charging and refill remain the project scope.

Make the complete system work

16. Mount and route cables

Put the battery and water tank low and secure them to the chassis. Mount the IR sensors rigidly, facing the route at the tested height; mount the ultrasonic sensor forward with a clear view. Keep nozzle aim fixed at the watering point. Keep hoses away from the wheels and route motor wires separately from analog sensor wires. Use strain relief and insulated terminals.

17. Bring functions together in this order

Step	Demonstration	Expected result
A	Dry / damp soil samples, no actuators	Correct raw readings and calibrated categories; invalid/stale readings inhibit watering.
B	IR sensors and ultrasonic only	Black/light transitions and measured distances are logged reliably.
C	Low-speed travel, pump physically disconnected	Line following and obstacle stop work. Missing echoes and lost line cause a stop, not continued travel.
D	Single-pot stopping marker	Robot repeatedly stops at the intended service point. A wide black marker needs route/state logic; two IR bits alone cannot identify an arbitrary pot.
E	Stationary robot, pump test into a cup	Calibrated volume within the chosen time limit; default/startup state is OFF.
F	One approved dry-soil job	Arrive, stop, dose, wait for absorption, read soil again; enforce maximum total dose/time.
G	Wet-soil request / obstacle / pause / lost link	No unwanted watering. Stop affected outputs; require a valid authorized state before resuming.
H	Repeat a previously handled job ID	No second dose from a duplicate message.

18. Record evidence for the sensor lab

Keep a log of timestamps, raw moisture ADC, both IR states, ultrasonic distance/validity, job ID, robot state and pump-on time. Measure actual delivered volume separately; pump time only estimates volume after calibration. Compare moisture readings against a defensible reference or characterize repeatability and error.

Your final code must include the required original C/C++, two explicit ISRs, two timer functions, ADC, communications and power saving. The supplied Uno diagnostics are deliberately not final firmware. At least one chosen sensor must differ from sensors used in your regular CEG2001 labs; confirm that with the group. [R8]

First session target: finish stages 2 and 3. You should have a working webcam, verified USB connections and raw readings from each identified sensor. Motor/battery commissioning starts only once the information on the next page has been filled in.

Bench record and source links

Ask the lab / supplier	Record before actuator and battery power-up
MSP430 + Booster	LaunchPad model: _____ Booster revision: _____
IR / soil / ultrasonic	Model and supply range: _____
Wheel motors	Rated voltage: _____ Running current: _____ Stall/start: _____
Dual motor driver	Model: _____ Per-channel continuous / peak current: _____
Water pump	Rated voltage/current: _____ Startup current: _____
Pump switch	Model, logic voltage, output terminal polarity: _____
Battery	Protected + / - / third wire functions: _____
Battery limits	Continuous/peak discharge; cutoff; charger model: _____
Regulators / fuse	Outputs/current ratings; fuse and wire ratings: _____

Reference documents

R1 Raspberry Pi: Pico series, official pinouts and Pico W power documentation

R2 Raspberry Pi: C/C++ SDK and VS Code setup

R3 DFRobot DFR0478 FireBeetle ESP32 documentation and board setup

R4 HC-SR04 datasheet hosted by SparkFun: 5 V supply and timing

R5 SunFounder: analogous 3.3-5 V IR proximity module; verify your own variant

R6 DFRobot DFR0457: power and logic limits; inductive-load protection

R7 Pololu TB6612FNG carrier: ratings for this specific carrier

R8 CEG2001/2002 Group Project Manual 2026, supplied PDF, pp. 2-5 and 7

R9 Unicell LI18650PB112W3: shortlisted pack; incomplete current/wiring specification

R10 Arduino Uno R3: power inputs and board specifications

Prepared from the supplied hardware photos and manufacturer documentation. Generic sensor examples do not identify an unmarked module. No physical wiring, firmware flashing, battery-load test or robot operation has been performed by this guide.